

that differ from those available to a high school graduate, but by going to school for extra months, they've missed an opportunity to earn in those months, so their lifetime earnings are actually lower than the earnings of people who enter the workforce directly after high school graduation and remain there.

Only after several years of college education does someone acquire skills that enable them to earn more over a lifetime than a high school graduate after we take into account the lost earning potential of the years spent at school. Then, college graduates (at 16 years of education) are at another earnings peak higher than the local high school peak. Once again, it's only a local one. Getting a little more education after earning a bachelor's degree leads to the same situation as getting a little more education after a high school diploma: you don't immediately acquire enough skills to compensate for the time not spent earning. Eventually, that's reversed, and you reach what looks like another peak after obtaining a postgraduate degree. It's hard to speculate much further beyond that, but this simplistic view of education and earnings will suffice for our purposes.

Climbing the Education Hill—the Right Way

For the individual we've imagined, drawn at 12.5 years of education on our graph, we can perform gradient ascent exactly as outlined previously.

[Listing 3-2](#) has a slightly altered version of the gradient ascent code we introduced in [Listing 3-1](#).

```
def income_derivative(edu_yrs):
    return(math.cos((edu_yrs - 10.6) * (2 * math.pi/4)) + 1/2)

threshold = 0.0001
maximum_iterations = 100000

current_education = 12.5
step_size = 0.001

keep_going = True
iterations = 0
while(keep_going):
    education_change = step_size * income_derivative(current_education)
    current_education = current_education + education_change
    if(abs(education_change) < threshold):
```